

Functions

ANSWER KEY



Function notation and evaluation

- 1 If $f(x) = 2x^2 - 3x + 1$, find $f(4)$.

$$f(4) = 2(16) - 3(4) + 1 = 32 - 12 + 1 = 21.$$

- 2 If $g(x) = \frac{x+3}{x-1}$, find $g(-2)$.

$$g(-2) = \frac{-2+3}{-2-1} = \frac{1}{-3} = -\frac{1}{3}.$$

Domain and range

- 3 Find the domain of $f(x) = \frac{1}{x-5}$.

The function is undefined when the denominator is zero: $x - 5 = 0$, so $x = 5$. Domain: all real numbers except $x = 5$.

- 4 Find the domain of $g(x) = \sqrt{x-3}$.

The expression under the square root must be non-negative: $x - 3 \geq 0$, so $x \geq 3$. Domain: $x \geq 3$.

Composite functions

- 5 If $f(x) = x^2 + 1$ and $g(x) = 2x - 3$, find $f(g(2))$.

First find $g(2) = 2(2) - 3 = 1$. Then $f(1) = 1^2 + 1 = 2$.

- 6 If $f(x) = 3x - 2$ and $g(x) = x + 4$, find $(g \circ f)(x)$ as a simplified expression.

$$(g \circ f)(x) = g(f(x)) = g(3x - 2) = (3x - 2) + 4 = 3x + 2.$$

Inverse functions

- 7 Find the inverse of $f(x) = 3x - 6$.

Let $y = 3x - 6$. Swap x and y : $x = 3y - 6$. Solve for y : $x + 6 = 3y$, so $y = \frac{x+6}{3}$. Thus $f^{-1}(x) = \frac{x+6}{3}$.

- 8 If $f(5) = 12$, find $f^{-1}(12)$.

By the definition of an inverse function, $f^{-1}(12) = 5$, since inverse functions reverse input and output pairs.

Transformations of functions

- 9 The graph of $y = f(x)$ is shifted 3 units right and 2 units up. Write the equation of the transformed function in terms of f .

Shifting right by 3 replaces x with $x - 3$; shifting up by 2 adds 2 to the output. Transformed function:
 $y = f(x - 3) + 2$.

- 10 Describe the transformation that takes $y = x^2$ to $y = -2(x + 1)^2 + 5$.

The graph is reflected over the x -axis and vertically stretched by a factor of 2 (from the -2), shifted 1 unit left (from $x + 1$), and shifted 5 units up (from $+5$).

Piecewise functions

- 11 A function is defined as $f(x) = x + 2$ for $x < 3$ and $f(x) = 2x - 1$ for $x \geq 3$. Find $f(1)$ and $f(5)$.

Since $1 < 3$, use the first piece: $f(1) = 1 + 2 = 3$. Since $5 \geq 3$, use the second piece: $f(5) = 2(5) - 1 = 9$.

Interpreting function graphs and increasing/decreasing behavior

- 12 A function's graph rises from $x = 0$ to $x = 3$, reaches a peak at $x = 3$, then falls from $x = 3$ to $x = 6$. State the intervals where the function is increasing and decreasing, and identify the type of point at $x = 3$.

The function is increasing on the interval $0 < x < 3$ and decreasing on the interval $3 < x < 6$. The point at $x = 3$ is a local maximum, since the function's value is greater there than at nearby points on both sides.

Modeling with function families

- 13 A company's profit is modeled by $P(x) = -2x^2 + 40x - 150$, where x is the number of units sold (in hundreds). Determine whether this is a linear, quadratic, or exponential model, and find the number of units that maximizes profit.

This is a quadratic model, identifiable by the x^2 term. The maximum occurs at $x = -\frac{b}{2a} = -\frac{40}{2(-2)} = 10$, meaning profit is maximized when 1000 units (10 hundreds) are sold.

Finding function values from a graph or table

- 14 A table shows: $f(0) = 3$, $f(1) = 5$, $f(2) = 9$, $f(3) = 15$. Determine whether this could represent a linear or quadratic function, by examining the differences between consecutive values.

First differences: $5 - 3 = 2$, $9 - 5 = 4$, $15 - 9 = 6$. Since the first differences are not constant (they increase by 2 each time), the function is not linear. Second differences: $4 - 2 = 2$, $6 - 4 = 2$ — these ARE constant, which is the signature of a quadratic function.

Zeros of a function and their graphical meaning

- 15 A function $f(x) = (x - 2)(x + 5)$ has zeros where it crosses the x -axis. Find the zeros, and state the y -intercept of the function.

Zeros occur where each factor equals zero: $x = 2$ and $x = -5$. The y -intercept is found at $x = 0$:
 $f(0) = (0 - 2)(0 + 5) = (-2)(5) = -10$, so the y -intercept is $(0, -10)$.

Even and odd functions

- 16 Determine whether $f(x) = x^2 - 3$ is even, odd, or neither, by evaluating $f(-x)$ and comparing to $f(x)$.

$f(-x) = (-x)^2 - 3 = x^2 - 3 = f(x)$. Since $f(-x) = f(x)$, the function is even (its graph is symmetric about the y -axis).

- 17 Determine whether $g(x) = x^3 - 2x$ is even, odd, or neither.

$g(-x) = (-x)^3 - 2(-x) = -x^3 + 2x = -(x^3 - 2x) = -g(x)$. Since $g(-x) = -g(x)$, the function is odd (its graph has point symmetry about the origin).

Modeling a real-world quantity with a function and evaluating it

- 18 The cost to rent a car is modeled by $C(d) = 35 + 0.20d$, where d is the number of miles driven. Find $C(150)$ and interpret the result in context.

$C(150) = 35 + 0.20(150) = 35 + 30 = 65$. This means renting the car and driving 150 miles costs 65 dollars total.

Solving for an input given an output

- 19 If $f(x) = 2x - 7$ and $f(a) = 15$, find the value of a .

$2a - 7 = 15$, so $2a = 22$, giving $a = 11$.

- 20 If $g(x) = x^2 - 4$ and $g(b) = 21$, find all possible values of b .

$$b^2 - 4 = 21, \text{ so } b^2 = 25, \text{ giving } b = 5 \text{ or } b = -5.$$

Multiple choice — easy

- 21 If $f(x) = 3x^2 - 5x + 4$, find $f(1)$.

a. 2 ✓

b. 4

c. -6

d. 0

$$f(1) = 3(1)^2 - 5(1) + 4 = 2.$$

- 22 If $f(x) = 3x^2 - 2x + 5$, find $f(6)$.

a. 101 ✓

b. 99

c. 103

d. 91

$$f(6) = 3(6)^2 - 2(6) + 5 = 101.$$

- 23 If $f(x) = -4x^2 + 4x + 7$, find $f(-6)$.

a. -161 ✓

b. -159

c. -175

d. -163

$$f(-6) = -4(-6)^2 + 4(-6) + 7 = -161.$$

24 If $f(x) = 4x^2 - 5x - 9$, find $f(4)$.

a. 35 ✓

b. 53

c. 33

d. 37

$$f(4) = 4(4)^2 - 5(4) - 9 = 35.$$

25 If $f(x) = -x^2 + 5x - 1$, find $f(5)$.

a. 4

b. -3

c. -1 ✓

d. 1

$$f(5) = -(5)^2 + 5(5) - 1 = -1.$$

26 If $f(x) = 2x^2 + 7x + 3$, find $f(4)$.

a. 57

b. 65

c. 63 ✓

d. 61

$$f(4) = 2(4)^2 + 7(4) + 3 = 63.$$

27 If $f(x) = -4x^2 + 3x + 1$, find $f(6)$.

a. -123

b. -125 ✓

c. -127

d. -120

$$f(6) = -4(6)^2 + 3(6) + 1 = -125.$$

28 If $f(x) = -x^2 - 5x + 1$, find $f(-1)$.

a. 10

b. 5 ✓

c. 3

d. 7

$$f(-1) = -(-1)^2 - 5(-1) + 1 = 5.$$

29 If $f(x) = 3x^2 + 4x + 3$, find $f(-3)$.

a. 18 ✓

b. 16

c. 12

d. 20

$$f(-3) = 3(-3)^2 + 4(-3) + 3 = 18.$$

30 If $f(x) = 3x^2 + 9x + 7$, find $f(-1)$.

a. -13

b. -1

c. 3

d. 1 ✓

$$f(-1) = 3(-1)^2 + 9(-1) + 7 = 1.$$

31 If $f(x) = -x^2 + 5x + 3$, find $f(5)$.

a. 1

b. 5

c. -3

d. 3 ✓

$$f(5) = -(5)^2 + 5(5) + 3 = 3.$$

32 If $f(x) = -x^2 - 4x + 2$, find $f(-3)$.

a. 3

b. 1

c. 7

d. 5 ✓

$$f(-3) = -(-3)^2 - 4(-3) + 2 = 5.$$

33 If $f(x) = -2x^2 + 7x + 5$, find $f(-2)$.

a. -17 ✓

b. -15

c. -19

d. -27

$$f(-2) = -2(-2)^2 + 7(-2) + 5 = -17.$$

34 If $f(x) = -x^2 - 5x - 3$, find $f(-5)$.

a. -5

b. 3

c. -3 ✓

d. -1

$$f(-5) = -(-5)^2 - 5(-5) - 3 = -3.$$

35 If $f(x) = 2x^2 + 2x + 9$, find $f(-6)$.

a. 67

b. 51

c. 69 ✓

d. 71

$$f(-6) = 2(-6)^2 + 2(-6) + 9 = 69.$$

Multiple choice — medium

36 If $f(x) = 2x + 7$ and $g(x) = -2x - 7$, find $f(g(-3))$.

a. 5 ✓

b. -1

c. 3

d. 7

First find $g(-3) = -2(-3) - 7 = -1$. Then $f(-1) = 2(-1) + 7 = 5$.

37 If $f(x) = -4x - 3$ and $g(x) = -3x + 5$, find $f(g(2))$.

a. 3

b. -2

c. -1

d. 1 ✓

First find $g(2) = -3(2) + 5 = -1$. Then $f(-1) = -4(-1) - 3 = 1$.

38 If $f(x) = x + 9$ and $g(x) = x - 1$, find $f(g(-6))$.

a. 2 ✓

b. -7

c. 4

d. 0

First find $g(-6) = (-6) - 1 = -7$. Then $f(-7) = (-7) + 9 = 2$.

39 If $f(x) = -4x + 6$ and $g(x) = -3x - 5$, find $f(g(2))$.

a. 48

b. 52

c. -11

d. 50 ✓

First find $g(2) = -3(2) - 5 = -11$. Then $f(-11) = -4(-11) + 6 = 50$.

40 If $f(x) = 2x + 3$ and $g(x) = -2x + 1$, find $f(g(-4))$.

a. 21 ✓

b. 9

c. 23

d. 19

First find $g(-4) = -2(-4) + 1 = 9$. Then $f(9) = 2(9) + 3 = 21$.

41 If $f(x) = -4x - 4$ and $g(x) = x - 7$, find $f(g(-1))$.

a. 28 ✓

b. -8

c. 26

d. 30

First find $g(-1) = (-1) - 7 = -8$. Then $f(-8) = -4(-8) - 4 = 28$.

42 If $f(x) = -3x - 5$ and $g(x) = 4x + 7$, find $f(g(4))$.

- a. 23
- b. -76
- c. -72
- d. -74 ✓

First find $g(4) = 4(4) + 7 = 23$. Then $f(23) = -3(23) - 5 = -74$.

43 If $f(x) = -4x - 1$ and $g(x) = x + 3$, find $f(g(6))$.

- a. -37 ✓
- b. -39
- c. -35
- d. 9

First find $g(6) = (6) + 3 = 9$. Then $f(9) = -4(9) - 1 = -37$.

44 If $f(x) = -x - 4$ and $g(x) = -2x + 8$, find $f(g(-5))$.

- a. -22 ✓
- b. -24
- c. 18
- d. -20

First find $g(-5) = -2(-5) + 8 = 18$. Then $f(18) = -(18) - 4 = -22$.

45 If $f(x) = -3x + 1$ and $g(x) = -3x + 4$, find $f(g(-2))$.

a. -27

b. -29 ✓

c. -31

d. 10

First find $g(-2) = -3(-2) + 4 = 10$. Then $f(10) = -3(10) + 1 = -29$.

46 If $f(x) = -4x - 9$ and $g(x) = 4x + 1$, find $f(g(1))$.

a. -31

b. 5

c. -27

d. -29 ✓

First find $g(1) = 4(1) + 1 = 5$. Then $f(5) = -4(5) - 9 = -29$.

47 If $f(x) = 3x - 3$ and $g(x) = 4x + 7$, find $f(g(6))$.

a. 31

b. 88

c. 90 ✓

d. 92

First find $g(6) = 4(6) + 7 = 31$. Then $f(31) = 3(31) - 3 = 90$.

48 If $f(x) = -2x - 4$ and $g(x) = 3x + 2$, find $f(g(-2))$.

a. 6

b. 4 ✓

c. -4

d. 2

First find $g(-2) = 3(-2) + 2 = -4$. Then $f(-4) = -2(-4) - 4 = 4$.

49 If $f(x) = -3x - 6$ and $g(x) = -2x + 3$, find $f(g(-2))$.

- a. 7
- b. -25
- c. -29
- d. -27 ✓

First find $g(-2) = -2(-2) + 3 = 7$. Then $f(7) = -3(7) - 6 = -27$.

50 If $f(x) = 4x + 5$ and $g(x) = -x - 2$, find $f(g(-1))$.

- a. -1
- b. 1 ✓
- c. 5
- d. 3

First find $g(-1) = -(-1) - 2 = -1$. Then $f(-1) = 4(-1) + 5 = 1$.

Multiple choice — hard

51 Find the inverse of $f(x) = -5x - 12$.

- a. $\frac{x + 14}{-5}$
- b. $\frac{x + 10}{-5}$
- c. $\frac{x + 12}{-5}$ ✓
- d. $\frac{x - 12}{-5}$

Let $y = -5x - 12$. Swap x and y : $x = -5y - 12$. Solving for y gives $f^{-1}(x) = \frac{x + 12}{-5}$.

52 Find the inverse of $f(x) = -3x - 10$.

- a. $\frac{x+10}{-3}$ ✓
- b. $\frac{x+8}{-3}$
- c. $\frac{x+12}{-3}$
- d. $\frac{x-10}{-3}$

Let $y = -3x - 10$. Swap x and y : $x = -3y - 10$. Solving for y gives $f^{-1}(x) = \frac{x+10}{-3}$.

53 Find the inverse of $f(x) = -4x + 1$.

- a. $\frac{x-1}{-3}$
- b. $\frac{x-3}{-4}$
- c. $\frac{x+1}{-4}$
- d. $\frac{x-1}{-4}$ ✓

Let $y = -4x + 1$. Swap x and y : $x = -4y + 1$. Solving for y gives $f^{-1}(x) = \frac{x-1}{-4}$.

54 Find the inverse of $f(x) = -5x + 12$.

- a. $\frac{x-14}{-5}$
- b. $\frac{x-10}{-5}$
- c. $\frac{x-12}{-5}$ ✓
- d. $\frac{x+12}{-5}$

Let $y = -5x + 12$. Swap x and y : $x = -5y + 12$. Solving for y gives $f^{-1}(x) = \frac{x-12}{-5}$.

55 Find the inverse of $f(x) = 5x + 9$.

- a. $\frac{x+9}{5}$
- b. $\frac{x-7}{5}$
- c. $\frac{x-11}{5}$
- d. $\frac{x-9}{5}$ ✓

Let $y = 5x + 9$. Swap x and y : $x = 5y + 9$. Solving for y gives $f^{-1}(x) = \frac{x-9}{5}$.

56 If $f(x) = -5x + 9$ and $f(t) = -36$, find the value of t .

a. 10

b. 8

c. 9 ✓

d. 11

$-5t + 9 = -36$. Solving: $-5t = -45$, so $t = 9$. Check: $f(9) = -5(9) + 9 = -36$ ✓.

57 If $f(x) = 5x - 4$ and $f(t) = -44$, find the value of t .

a. -8 ✓

b. -6

c. -9

d. -7

$5t - 4 = -44$. Solving: $5t = -40$, so $t = -8$. Check: $f(-8) = 5(-8) - 4 = -44$ ✓.

58 If $f(x) = -3x + 6$ and $f(t) = -18$, find the value of t .

a. 7

b. 10

c. 9

d. 8 ✓

$-3t + 6 = -18$. Solving: $-3t = -24$, so $t = 8$. Check: $f(8) = -3(8) + 6 = -18$ ✓.

59 If $f(x) = -2x + 6$ and $f(t) = 8$, find the value of t .

- a. 1
- b. -2
- c. -1 ✓
- d. 0

$-2t + 6 = 8$. Solving: $-2t = 2$, so $t = -1$. Check: $f(-1) = -2(-1) + 6 = 8$ ✓.

60 If $f(x) = 2x$ and $f(t) = -14$, find the value of t .

- a. -6
- b. -5
- c. -7 ✓
- d. -8

$2t = -14$, so $t = -7$. Check: $f(-7) = 2(-7) = -14$ ✓.

61 Which value of x is NOT in the domain of $f(x) = \frac{1}{x+8}$?

- a. -9
- b. -8 ✓
- c. -7
- d. 8

The function is undefined when the denominator is zero: $x + 8 = 0$, so $x = -8$. This is the excluded value.

62 What is the smallest value of x in the domain of $f(x) = \sqrt{x + 4}$?

- a. -3
- b. -5
- c. 4
- d. -4 ✓

The expression under the square root must be non-negative: $x + 4 \geq 0$, so $x \geq -4$. The smallest such value is $x = -4$.

63 Which value of x is NOT in the domain of $f(x) = \frac{1}{x - 2}$?

- a. 2 ✓
- b. -2
- c. 1
- d. 3

The function is undefined when the denominator is zero: $x - 2 = 0$, so $x = 2$. This is the excluded value.

64 What is the smallest value of x in the domain of $f(x) = \sqrt{x - 9}$?

- a. 8
- b. -9
- c. 9 ✓
- d. 10

The expression under the square root must be non-negative: $x - 9 \geq 0$, so $x \geq 9$. The smallest such value is $x = 9$.

65 What is the smallest value of x in the domain of $f(x) = \sqrt{x - 7}$?

- a. -7
- b. 8
- c. 6
- d. 7 ✓

The expression under the square root must be non-negative: $x - 7 \geq 0$, so $x \geq 7$. The smallest such value is $x = 7$.